

File Sharing Using Access Control

Project Proposal Documentation

1. Introduction

This project proposes a secure file-sharing system that uses access control mechanisms to ensure that only authorized users can access, modify, or share files.

2. Problem Statement

Many file-sharing systems do not provide sufficient control over who can access sensitive information. This project addresses that issue through role-based access control.

3. Objectives

- Develop a secure file-sharing platform.
- Implement authentication and authorization.
- Provide role-based access control (RBAC).
- Allow file owners to manage permissions.
- Maintain activity logs.

4. Proposed System

The system enables users to upload, share, and download files securely. Permissions are checked before access is granted.

5. User Roles

Admin: Manages users and settings.

Owner: Uploads files and controls sharing permissions.

User: Accesses files according to assigned permissions.

6. Technologies

Frontend: HTML, CSS, JavaScript

Backend: PHP/Python/Node.js

Database: MySQL

Security: Authentication, Encryption, ACL/RBAC

7. Expected Outcomes

Improved security, controlled access to files, enhanced privacy, and efficient permission management.

8. Conclusion

The proposed system provides a secure and practical solution for file sharing through access control mechanisms.